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Morphometric trait analysis and machine learning-based yield modeling in wood apple (*Feronia limonia* L.)

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Abstract

Background Wood apple is a hardy yet underutilized fruit tree of the Indian subcontinent, valued for its nutritional, medicinal, and ecological significance. Despite its potential as a climate-resilient fruit species, the determinants of yield variability remain poorly characterized. This study aimed to quantify how morphometric descriptors of canopy architecture, floral, and fruit traits explain yield variation across 62 wood apple genotypes. By integrating multivariate statistics with explainable machine-learning models (Random Forest + SHAP), we provide the first data-driven framework for identifying trait combinations that govern productivity in this underutilized tree species. The approach offers a novel, interpretable path toward ideotype selection and precision orchard design.

Results Extensive morphometric variability was observed across the 62 genotypes for vegetative, foliar, floral, fruit and seed traits, indicating a broad genetic base. Yield per tree ranged widely from 35 to 127 kg, with a mean of 75 kg tree^{-1}. Principal Component Analysis (PCA) showed that canopy architecture, branch traits, and leaf–fruit attributes collectively explained 31.1% of the total variation. Correlation analysis revealed positive associations of yield with tree shape, pulp colour, and fruit-bearing tendency, whereas ornamental fruit traits and excessive spine density were negatively related. The optimized Random Forest (RF) model achieved strong predictive performance on the test dataset (R^2 = 0.84; RMSE = 9.45 kg; MAE = 7.12 kg), significantly outperforming Multiple Linear Regression (R^2 = 0.62), Support Vector Regression (R^2 = 0.76), and the Deep Learning (MLP) model (R^2 = 0.71). RF identified tree shape (16%), open flower colour (11.3%), and pulp colour (9.0%) as the most influential predictors of yield. SHAP analysis further clarified the non-linear and interactive effects among traits, highlighting the combined influence of canopy vigour, reproductive efficiency, and fruit-quality attributes on productivity. Hierarchical clustering grouped the genotypes into three clusters, with Cluster 2 characterized by compact canopies, superior reproductive traits, and desirable pulp features showing the highest and most stable yield (mean 84.6 kg tree^{-1}). Cluster 0 displayed moderate-to-high yields (79.7 kg tree^{-1}) but with greater variability, while Cluster 1 comprised the lowest-yielding genotypes (70.4 kg tree^{-1}).

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These findings confirm that productivity in wood apple is jointly regulated by architectural and reproductive traits through coordinated source–sink dynamics.

Conclusions Wood apple yield is governed by an integrated suite of architectural and reproductive traits, rather than single descriptors. Genotypes with compact canopies, regular bearing habit, and consumer-preferred pulp characteristics emerge as promising ideotypes for high productivity and orchard efficiency. By combining Random Forest and SHAP, this study demonstrates the practical value of explainable machine-learning tools in identifying actionable trait combinations and providing a robust, trait-based framework to support data-driven breeding and climate-smart orchard design in underutilized perennial fruit crops.

Keywords Wood apple, Morphometric traits, Yield prediction, Machine learning, PCA, SHAP, Random forest

Introduction

Fruit crops play a vital role in global horticulture, contributing to nutritional security, rural livelihoods, and value-added agro-industries [1, 2]. Wood apple (*Feronia limonia* L.; Rutaceae), also known as elephant apple, is a hardy perennial fruit tree native to the Indian subcontinent and Southeast Asia. Its resilience to drought, heat, and nutrient-poor soils makes it particularly suitable for semi-arid and climate-stressed production systems [3–5]. The fruit, enclosed in a hard shell and rich in carbohydrates, minerals and bioactive compounds, is widely used in beverages, confectionery, and traditional medicinal preparations [6–8]. Although national production statistics remain limited, improved orchards can yield 20–30 t ha^{-1} under appropriate management [9]. In Gujarat alone, the reported area and production are 233 ha and 1,645 metric tonnes respectively [10]. Despite modest aggregate production, wood apple supports local markets, cottage-scale processing, and smallholder income streams, underscoring the need for genetic improvement and enhanced yield potential [11–13].

Yield prediction in perennial fruit crops is inherently complex due to extended juvenile phases, irregular flowering, alternate bearing, and sensitivity to environmental fluctuations [14, 15]. These biological challenges render conventional statistical approaches inadequate for underutilized crops such as wood apple. Traditional analytical approaches such as ANOVA-based trait evaluation, correlation analysis, and linear regression have been useful for characterizing variability but are limited in capturing non-linear interactions and multicollinearity among morphometric traits. These constraints are particularly restrictive for underutilized species like wood apple. Recent advances in artificial intelligence (AI) and machine learning (ML) offer new opportunities to model complex, high-dimensional trait–yield relationships with improved accuracy and interpretability [16, 17]. Moreover, ML-based trait analysis has been successfully applied in diverse plant systems to predict stress responses and yield patterns from complex morphological and physiological datasets, demonstrating its broader utility for data-driven horticulture [18–25].

Morphometric descriptors including canopy architecture, branching pattern, leaf morphology, fruit size, pulp characteristics, floral traits, and seed morphology represent integrative indicators of vegetative vigour, reproductive efficiency, and fruit quality. Floral traits such as colour, inflorescence position and flowering pattern influence pollinator visitation and fruit set in this cross-pollinated species, while seed traits reflect developmental processes that shape sink strength and fruit maturation [26]. Similar trait-yield associations have been reported in grape, guava, and mango where canopy volume, stem girth, leaf metrics and fruit dimensions act as robust predictors of productivity [27–29]. Multivariate tools such as PCA, regression modelling, and modern ML techniques including Random Forests and neural networks have further enhanced the ability to detect meaningful interactions in perennial crops [19, 30, 31].

Despite these advances, there remains no comprehensive morphometric–yield dataset for wood apple, and explainable ML approaches (RF + SHAP) have not yet been applied to identify key trait determinants or ideotypes in this species. Therefore, a combined PCA–Random Forest–SHAP framework was implemented for dimensionality reduction, nonlinear prediction, and model interpretability, respectively. This study addresses this gap by integrating large-scale morphometric characterization with machine learning and SHAP-based interpretability to quantify trait–yield relationships. A trait-driven yield-prediction pipeline is developed for this underutilized fruit tree, enabling deeper insight into complex, non-linear interactions and supporting ideotype identification. By combining hierarchical clustering with SHAP interpretation, the study reveals biologically meaningful genotype groups and the underlying drivers of productivity. Against this backdrop, the present study was designed to: (i) identify morphometric traits significantly associated with yield in wood apple, (ii) apply ML-based models to predict yield from trait datasets, and (iii) classify genotypes into performance groups using clustering approaches. The outcomes are expected to provide novel insights into trait yield relationships, support breeding efforts for compact and high-yielding ideotypes, and contribute to precision and climate-smart horticultural management of underutilized perennial fruit crops.

Materials and methods

Experimental site

The study was carried out at the Central Horticultural Experiment Station (CHES), ICAR, Vejalpur, Panchmahal, Gujarat, India (22°41′N, 73°33′E; altitude 113 m above sea level) over two consecutive fruiting seasons (2023 and 2024). The experimental site is characterized by a hot, semi-arid climate, with an average annual rainfall of 575 mm. Mean annual temperatures vary between 18.2 and 36.8 °C, while peak summer temperatures can rise to 44 °C. Climatic variables were intentionally excluded from the present analysis because all genotypes were evaluated at a single experimental location under a uniform agro-climatic regime, where limited temporal and spatial variability would confound rather than enhance trait–yield modelling. Nevertheless, the integration of multi-season climatic covariates is recognized as a key priority for future multi-environment validation and model generalization. The soils of the experimental farm are predominantly shallow, with a clay-loam texture, a pH of approximately 6.8, and organic carbon content ranging from 0.45 to 0.50%.

Plant material

The study involved 62 wood apple genotypes maintained in the field gene repository at CHES, Panchmahal, Gujarat, India. The plant materials were collected from natural populations located in different regions with geographical coordinates in 2015-16 (Supplementary Table 1). The collected plant materials were identified by Dr. Vikas Yadav. A voucher specimens of these materials have been deposited in the publicly available herbarium of ICAR-Central Horticultural Experiment Station, Vejalpur with deposition number PD-3430. Each genotype, represented by trees aged 9–10 years, was planted in a rectangular layout with row and intra-row spacing of 10 × 5 m. The experiment was laid out in a Randomized Complete Block Design (RCBD) with three replications. Recommended crop management practices were implemented to ensure optimal tree growth [32]. To evaluate phenotypic variation among the genotypes, 31 morphometric traits were assessed (Fig. 1) and coded (Supplementary Table 2) following standard protocols [33–35].

Fig. 1 Representative leaf, flower, fruit, pulp, seed and tree morphology of the studied wood apple accessions

Morphological assessment of tree, leaf, and fruit traits

Morphological variation among 62 wood apple genotypes was evaluated using 31 traits, covering vegetative, leaf, floral, fruit, and seed characteristics, including tree growth habit (TGH), tree shape (TS), trunk bark color (TBC), branch density (DB), branch angle (BA), vegetative life cycle (VLC), new flush color (NFC), leaf size (LS), leaf petiole color (LPC), leaflet size (LtS), lamina attachment (LtLA), petiolule color (LtPlC), leaflet color (ventral: LtCV; dorsal: LtCD), lamina shape (LtLS), lamina apex shape (LtLAS), petiole wings (PW), spine length (SL), spine density per meter shoot (SDPS), color of open flower (COF), fruit shape (FS), fruit surface texture (FST), fruit color (FC), fruit stem end (FStE), fruit styler end (FSyE), pedicel color (PdC), tree-bearing tendency (TBT), fruit-bearing habits (FBT), pulp color (PC), seed shape (SS), and seed color (SC). Observations were recorded following standard descriptors developed for *Citrus* [33] and related species [34, 35].

The tree traits (TGH, TS, TBC, DB, and VLC) were measured before fruit harvest in the second week of October. Fully mature leaves (4–6 months old) from the middle portion of the current season's growth were collected for leaf and leaflet observations ($n = 50$). Flower traits were recorded from samples collected during February-March ($n = 30$), and fruit traits were measured from fruits harvested between late October and November ($n = 10$) (IPGRI, 1999). All samples were collected from 2 to 3 representative trees per genotype. Fruit yield per tree was determined by summing the fruits harvested during the fruiting season, from the last week of October to November.

Machine learning framework

A comprehensive data-analysis pipeline was developed to evaluate morphometric variation and model yield using machine learning (ML) approaches. The dataset consisted of 62 wood apple genotypes, each characterized for 31 qualitative and 18 quantitative traits. Trait descriptors were coded using standardized IPGRI-based criteria [33], and all measurements were recorded with three biological replications per genotype. A complete description of the qualitative coding scheme is provided in Supplementary Table 3. Before modeling, the dataset was examined for missing values, measurement inconsistencies, and extreme outliers. Continuous variables were standardized to zero mean and unit variance. Outliers were identified using both the interquartile range (IQR) rule and multivariate Mahalanobis distance, with biologically implausible values removed or corrected. The cleaned dataset was split into training (80%) and testing (20%) subsets, maintaining trait distribution balance. Explainability was achieved using SHAP (SHapley Additive exPlanations) values, enabling trait-wise attribution of yield predictions. SHAP outputs were integrated with k-means clustering to derive biologically meaningful ideotypes, highlighting key architectural and reproductive traits associated with high yield. All computations were performed on a workstation equipped with Intel® Core™ i7-12700 CPU, 32 GB RAM, and NVIDIA RTX 3060 GPU (12 GB). Random Forest hyperparameters included: n_estimators = 100, max_depth = None, minimum samples split = 2, and random_state = 42. To ensure the robustness of the yield prediction model, the Random Forest (RF) regressor was benchmarked against three alternative algorithms: Multiple Linear Regression (MLR) as a baseline, Support Vector Regression (SVR) (kernel='rbf', C = 1.0, epsilon = 0.1), and a Deep Learning Multilayer Perceptron (MLP). The MLP architecture consisted of two hidden layers (64 and 32 neurons) using ReLU activation and an Adam optimizer.

Hyperparameter optimization for the RF model was conducted using a Grid Search with 5-fold cross-validation. The search space included n_estimators (50, 100, 200), max_depth (None, 10, 20), and min_samples_split (2, 5, 10). The optimal configuration selected was n_estimators = 100, max_depth = 12, min_samples_split = 5, and min_samples_leaf = 2. Model performance was evaluated on a held-out test set (20%) using Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

Statistical analyses

The experiment was conducted during the years 2023 and 2024 fruiting season using a randomized complete block design (RCBD). All data analyses were conducted using Python (version 3.9) with a suite of libraries tailored for statistical and machine learning tasks: pandas for data manipulation [36], NumPy for numerical computations [37], scikit-learn for machine learning and preprocessing [38], seaborn and matplotlib for visualization [39, 40], and SHAP for model interpretability [41]. The dataset comprising 186 observations of 62 wood apple genotypes (three replications each), included 31 morphometric traits and yield (YT). Data were split into training (80%) and testing (20%) sets with a random seed of 42, and traits were standardized using StandardScaler to normalize scales prior to analysis [38]. Five complementary analyses were performed to investigate trait-yield relationships. First, Pearson's correlation coefficients were calculated and visualized as a half-triangle heatmap to assess linear associations between traits and yield [42]. Second, Principal Component Analysis (PCA) reduced the 32 traits to two components, with variance explained and trait loadings examined [43]. Third, a Random Forest Regressor (100 trees, random seed 42) modeled yield, with performance evaluated via Mean Squared Error and R^2, and feature importance extracted [44]. Fourth, SHAP values from the Random Forest were computed to interpret trait contributions to predictions [41]. Genotypes were classified into performance groups using K-Means clustering. The optimal number of clusters (k = 3) was determined using the Elbow Method and validated via Silhouette Analysis, which yielded a coefficient of 0.56, indicating a high degree of separation between clusters. Euclidean distance was used as the similarity metric [45].

Result

Frequency distribution of morphometric traits

The frequency distribution analysis revealed wide morphological variability among the 62 wood apple genotypes across vegetative, foliar, floral, fruit, and seed descriptors, underscoring a strong genetic base (Fig. 2). Most trees were erect (32) with ellipsoid (31) or obloid (17) canopies, while bark colour was predominantly blackish-grey (34). Branching was mostly dense (32) with narrow to medium angles. A majority were deciduous (40), with reddish-green new flushes (40), suggesting seasonally adaptive growth forms. Leaf traits demonstrated diversity, with medium-sized leaves (25), sessile attachment (55), and green laminae (52) being most common. Ovate (24) and obovate (12) lamina shapes dominated, often with acuminate apices (24). Petiole wings were absent in most genotypes (56). Spine traits were relatively uniform, with medium length (26) and low density (26). Floral characters exhibited pronounced variability, with flower colours ranging from light purple (16) and yellow (15) to purple (9). Most genotypes bore flowers terminally and laterally (50), in bunches (50), indicating reproductive plasticity favourable for cross-pollination. Fruit descriptors were highly polymorphic, with round fruits (34), rough surfaces (38), and external colours varying from dull white (27) to greyish (20). Stem ends were curved or convex, styler ends round or concave, and pedicels mostly greyish (31). Pulp colour, a key consumer attribute, ranged from pale gold (18) to creamy white (12). Seed traits further confirmed heterogeneity, with broad ovate (24) and oblong (14) shapes, and seed coat colours mostly brownish white (30) or brown (22).

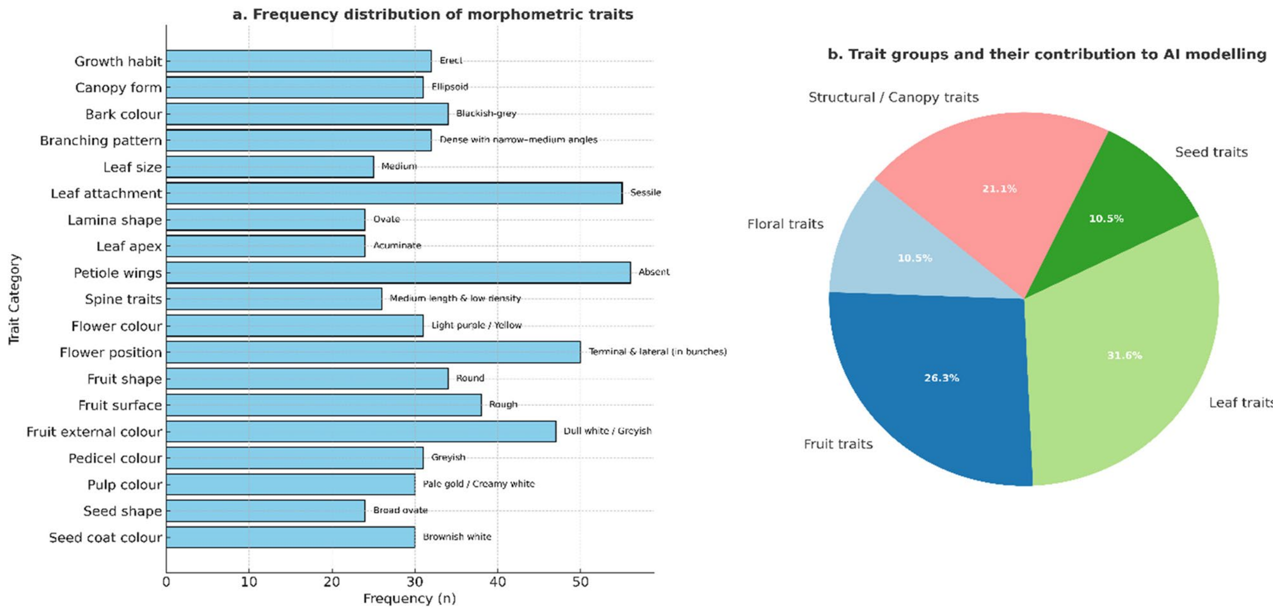

Fig. 2 Frequency distribution of morphometric traits and their grouped contribution to AI-based modelling in 62 wood apple genotypes. (a) The bar chart illustrates the frequency distribution of key qualitative morphometric traits related to canopy architecture, floral, fruit, and seed characteristics. Traits such as erect growth habit, ellipsoid canopy form, ovate lamina shape, and pale-gold pulp color were predominant among genotypes. (b) The pie chart depicts the proportional contribution of different trait categories to AI modelling, where leaf traits (31.6%) and fruit traits (26.3%) accounted for the largest share, followed by structural/canopy (21.1%), floral (10.5%), and seed traits (10.5%). This highlights the predominance of vegetative and fruit-related parameters in determining model performance and phenotypic variability

Correlation analysis

The correlation analysis provided critical insights into the interdependencies among vegetative, reproductive, and yield-related traits in the studied 62 genotypes (Fig. 3). From an AI and modeling perspective, such correlations are essential for feature engineering, as they determine which variables can serve as reliable predictors of yield or quality in supervised learning pipelines. A moderate positive correlation was observed between yield (YT) and tree shape (TS, $r=0.35$), pulp colour (PC, $r=0.23$), fruit bearing habit (FBT, $r=0.23$), and tree bearing tendency (TBT, $r=0.22$). These associations suggest that genotypes with vigorous architecture, regular bearing capacity, and desirable pulp colour tend to be more productive. Importantly, this validates the predictive utility of structural and reproductive descriptors in machine learning models, where they can be prioritized as high-weight features during yield prediction. The positive correlation of YT with PC also highlights the potential for multi-objective modeling, where consumer-preferred pulp traits contribute simultaneously to both productivity and marketability. Such dual-purpose features are particularly valuable in multi-output regression models, enabling the simultaneous optimization of yield and quality indices. By contrast, YT exhibited weak to negative correlations with ornamental fruit quality traits such as fruit surface texture (FST, $r=-0.14$), fruit colour (FC, $r=-0.14$), and seed colour (SC, $r=-0.03$). This pattern reflects a resource-allocation trade-off, where investment in external fruit attributes compromises yield potential. In AI-based selection pipelines, such negative associations can be encoded as penalty functions or trade-off constraints, ensuring that breeding decisions do not inadvertently prioritize market aesthetics at the expense of productivity. Vegetative traits displayed strong intercorrelations that carry direct implications for predictive modeling. Tree growth habit (TGH) was strongly correlated with branching angle (BA, $r=0.91$) and new flush colour (NFC, $r=0.86$), indicating a coordinated growth module that could serve as a latent feature cluster in dimensionality-reduction approaches. Conversely, the negative correlation of TGH with branching density (DB, $r=-0.95$) suggests an architectural balance where taller trees bear fewer but stronger branches. While TGH itself showed only a weak correlation with yield ($r=0.14$), its indirect contribution through canopy ventilation, light interception, and disease resistance highlights the importance of including such traits in explainable AI frameworks (e.g., SHAP values) to capture non-linear and indirect effects. Reproductive and fruit-related traits also showed coordinated associations. For example, fruit shape (FS) correlated with fruit stem-end (FStE, $r=0.22$) and pedicel colour (PdC, $r=0.40$), while fruit colour (FC) correlated strongly with pedicel colour ($r=0.62$). Such clustering of traits indicates genetically linked modules that can be exploited in unsupervised learning tasks such

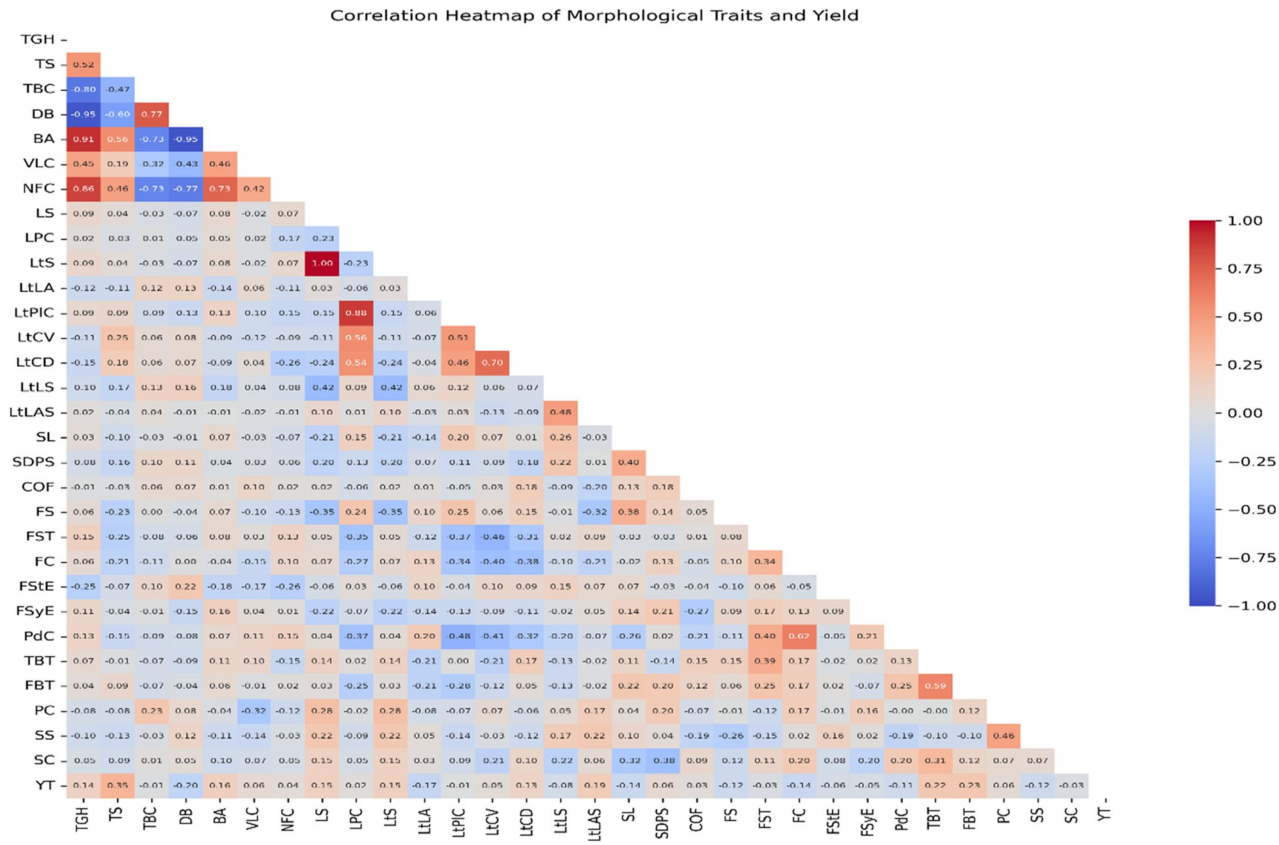

Fig. 3 Correlation heatmap analysis of 31 morphometric traits of 62 wood apple genotypes. The matrix depicts pairwise Pearson correlation coefficients between 31 morphometric traits. Red and blue shades indicate positive and negative correlations, respectively; darker colors represent stronger correlations. Abbreviations- TGH: Tree Growth Habit (m), TS: Tree shape, TBC: Tree bark colour, DB: density of branch, BA: branch angle: VLC: Vegetative life cycle, NFC: New flush colour, LS: Leaf Size, LPC: Leaf petiole colour, LtS: Leaflet Size, LtLA: Leaflet lamina attachment, LtPIC: leaflet petiolulecolour, LtCV: leaflet colour ventral side, LtCD: leaflet colour dorsal side, LtLS: leaflet lamina shape, LtLAS: leaflet lamina apex shape, SL: Spine Length (cm), SDPS: Spine density per meter shoot, COF: colour of open flower, FS: Fruit shape, FST: Fruit surface texture, FC: Fruit colour, FStE: Fruit stem end, FSyE: fruit styler end, PdC: Pedicel colour, TBT: Tree bearing tendency, FBT: fruit bearing habits, PC: Pulp colour, SS: Seed shape, SC: Seed colour, YT: Yield/ Tree

as clustering or principal component analysis (PCA). These associations suggest that indirect selection for one trait (e.g., pedicel pigmentation) could simultaneously enhance related traits, improving the predictive stability of trait groups in machine learning pipelines.

Principal component analysis (PCA)

Principal Component Analysis (PCA) was performed to assess the contribution of morphological traits toward yield/tree (YT) in the studied genotypes. The first two principal components (PC1 and PC2) explained 17.8% and 13.3% of the total variance, respectively, cumulatively accounting for 31.1% of the variability. The distribution of genotypes in the PCA scatterplot revealed considerable diversity in morphometric traits, as reflected by the wide spread of accessions along both axes (Fig. 4). Genotypes clustered across PC1 and PC2 based on trait combinations such as tree growth habit (TGH), tree shape (TS), branching density (DB), branch angle (BA), and vegetative life cycle (VLC), which were the major contributors to variation. Leaf-related descriptors including leaf size (LS), leaflet size (LtS), leaflet lamina attachment (LtLA), leaflet colour (LtCV, LtCD), and leaflet lamina apex shape (LtLAS) also played a role in differentiation, indicating that both vegetative and reproductive morphology influenced the separation of genotypes. Fruit related traits such as fruit shape (FS), fruit surface texture (FST), pulp colour (PC), and seed shape (SS) further explained yield-related diversity.

Colour coding by yield (YT) revealed that high-yielding genotypes represented in yellow–green shades were distributed predominantly towards the positive side of PC1, indicating that traits aligned along this axis had stronger association with yield. Genotypes with larger tree stature (TGH, TS), moderate branch density (DB), wider branch angles (BA), and balanced fruiting traits (FS, PC, SC) tended to occupy this space. Conversely, genotypes clustered towards the negative side of PC1 or lower quadrants, often characterized by smaller leaf and fruit traits, lower spine density (SDPS), and less desirable reproductive features, corresponded with relatively lower

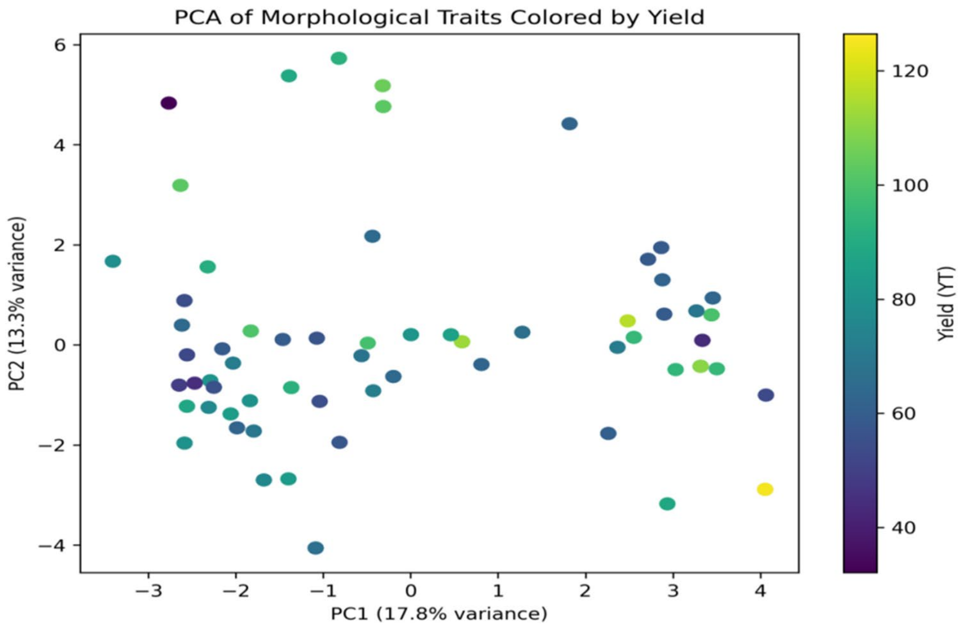

Fig. 4 The PCA-biplot displays the morphometric-yield relationship among 62 genotypes. The plot illus-trates the distribution of genotypes based on the first two principal components (PC1: 17.8% var-iance, PC2: 13.3% variance), summarizing the variation in morphological traits. Each point repre-sents an individual genotype, color-coded according to yield (YT). Genotypes with higher yield values appear in yellow to green shades, whereas lower-yielding ones are represented by darker blue to purple shades. The dispersed pattern indicates substantial diversity among genotypes, re-flecting multivariate differentiation in morphometric attributes associated with yield potential.

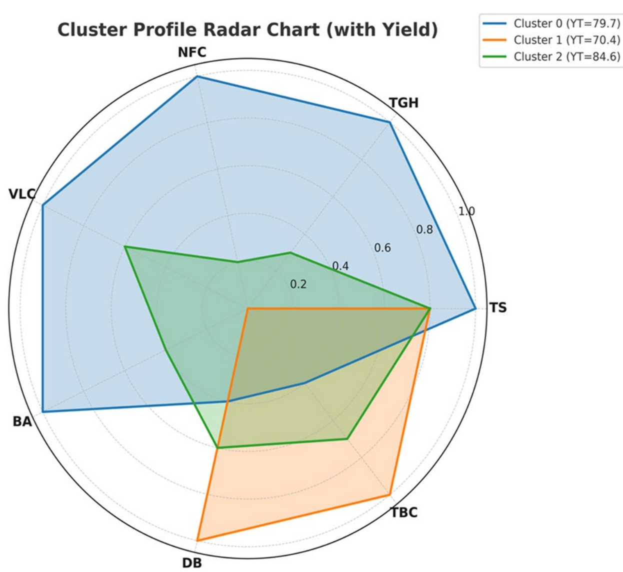

Fig. 5 Radar plot illustrating the mean performance of 31 morphometric and yield-related traits across three clusters of 62 wood apple genotypes. Each cluster (Cluster 0, Cluster 1, and Cluster 2) represents distinct phe-notypic profiles, with yield (YT) values indicated in parentheses. Cluster 2 (YT = 84.6) exhibited superior performance for key fruit and yield traits, whereas Cluster 1 (YT = 70.4) showed comparatively lower trait values. Cluster 0 (YT = 79.7) displayed intermediate to high performance across most parameters. For abbreviations see Fig. 3

yields (purple–blue shades).The dispersion of genotypes suggests that yield is not governed by a single trait but rather by a composite effect of vegetative vigor, reproductive attributes, and fruit quality. For instance, accessions with larger fruits (FS, FC), attractive pulp colour (PC), and moderate seed size (SS, SC) coupled with robust vegetative growth exhibited higher yield potential. On the other hand, traits like excessive spine density (SDPS), smaller leaf size (LS, LtS), or narrow leaflet shapes (LtLS) were generally associated with reduced yield. Overall, PCA analysis provided valuable insights into the relationship between morphometric traits and yield, highlighting that a balance between tree vigor, leaf morphology, and fruit quality traits is critical for maximizing productivity. These findings can be effectively utilized for selecting promising genotypes with desirable combinations of traits for breeding and orchard establishment.

Clustering analyses

The clustering of wood apple genotypes revealed three distinct Clusters with clear differences in morphometric and reproductive traits, which were strongly associated with yield performance (Figs.5 and 6; Supplementary Table 4). Genotypes grouped in Cluster 0 were characterized by the highest tree vigour and balanced canopy traits, with moderate fruit bearing habit and pulp colour, coupled with relatively higher fruit shape and styler end. These features contributed to a mean yield of 79.7 kg/tree, indicating that taller and vigorous trees

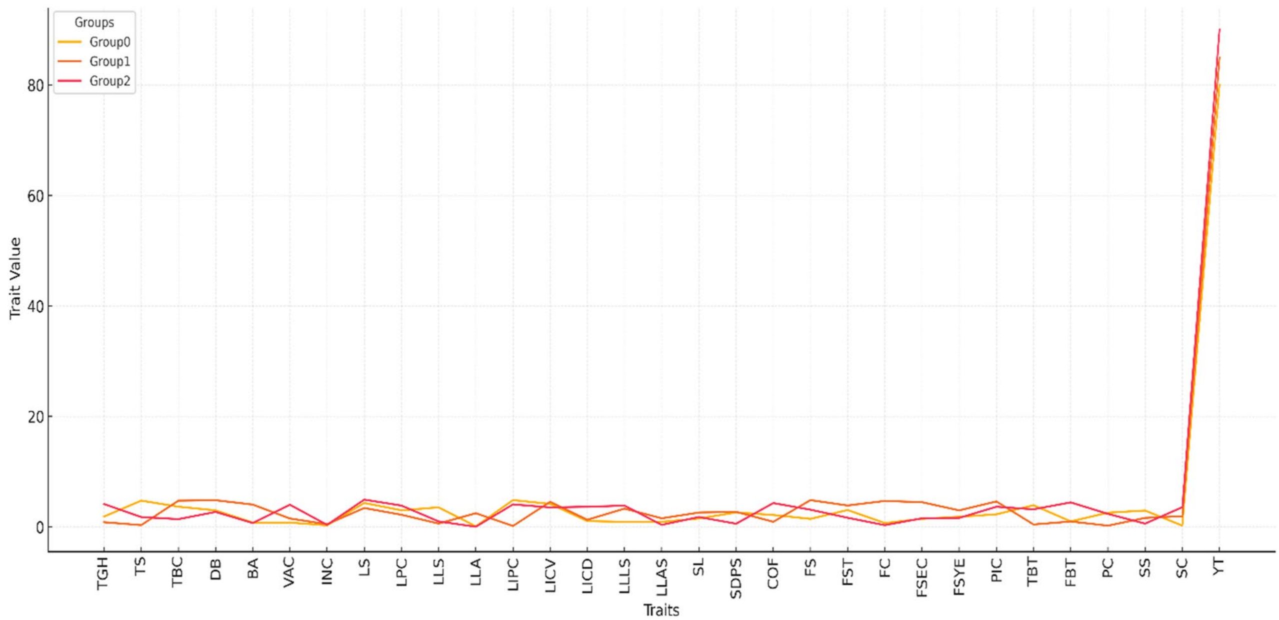

Fig. 6 Parallel coordinates plot showing the mean performance of 31 morphometric and yield traits across three clusters of 62 wood apple genotypes. Each colored line represents a cluster (Cluster 0, Cluster 1, and Cluster 2), highlighting trait-wise variation patterns among clusters. The plot reveals considerable inter-cluster variability across several morphological and yield-contributing traits, with distinct differences in yield (YT) and associated parameters. This visualization emphasizes the multivariate differentiation and potential trait associations contributing to yield performance among genotypes. For abbreviations see Fig. 3.

with balanced reproductive traits can achieve satisfactory productivity. By contrast, Cluster 1 comprises genotypes with well-developed leaf and floral descriptors, as reflected in higher LtLS, FS, and COF. However, this cluster showed a relatively lower yield (70.4 kg/tree), which can be attributed to reduced tree growth habit and tree bearing tendency. Interestingly, Cluster 2 genotypes were relatively compact but displayed superior reproductive traits such as fruit shape, colour of open flower, and fruit styler end. This combination supported the highest mean yield of 84.6 kg/tree, highlighting that compact architecture coupled with strong reproductive efficiency is more advantageous for productivity than excessive vegetative growth. These findings demonstrate that genotypes from Cluster 2 represent the most promising ideotypes for breeding compact, high-yielding trees, whereas Cluster 0 can be used to introgressive vigour-related traits, and Cluster 1 is valuable for studying floral and leaf attributes, though it requires improvement in bearing stability.

The parallel coordinates plot (Fig. 6) revealed distinct patterns of variation across morphological, and reproductive traits in the evaluated genotypes. Most traits, including tree growth habit (TGH), tree shape (TS), branch angle (BA), and fruit characteristics (PC, FBT), exhibited relatively low variability across the three Clusters (coded as 0, 1, and 2). In contrast, yield trait (YT) displayed pronounced divergence, with sharp increases in trait value compared to other descriptors.

The yield distribution across clusters

The yield distribution across clusters, as illustrated in the boxplot (Fig. 7), further substantiates the Clustering patterns obtained from hierarchical clustering. Cluster 0 exhibited a wide yield variation ranging from 35 to 127 kg/tree, with a median around 70 kg/tree. This high variability indicates that while some genotypes within this group have strong yield potential, others remain relatively unproductive, reflecting the influence of excessive vegetative vigour and unstable reproductive efficiency. Cluster 1, on the other hand, demonstrated more consistent yield performance but at comparatively lower levels, with a narrower range (40–100 kg/tree) and median yield also near 70 kg/tree. The stability of Cluster 1 suggests uniformity in genetic expression for floral and fruit traits, but its lower productivity aligns with weaker bearing ability observed in the trait analysis. In contrast, Cluster 2 showed the highest and most stable yield performance, with yields tightly distributed between 72 and 105 kg/tree and a median close to 90 kg/tree. This compact distribution confirms that genotypes in Cluster 2 are not only high-yielding but also stable across replications, owing to their favorable balance of compact tree architecture and superior reproductive traits. The presence of only a few outliers in Cluster 2 further highlights its robustness, whereas the wider spread in Cluster 0 suggests the need for careful selection of promising individuals.

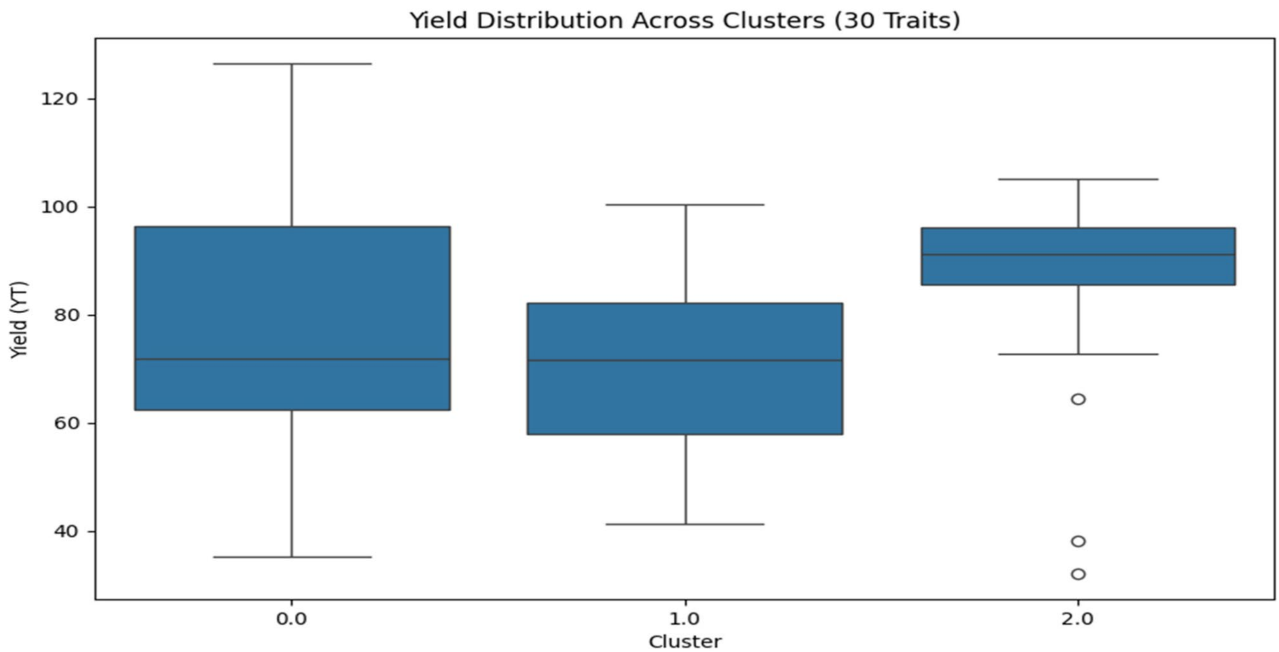

Fig. 7 Box plot illustrating yield (YT) distribution across three clusters of 62 wood apple genotypes. The plot shows clear variation in yield performance among clusters, with Cluster 2 exhibiting the highest median yield and least variability, while Cluster 1 recorded the lowest yield. Cluster 0 displayed a wide yield range, indicating greater within-group diversity. These differences highlight distinct yield potential patterns among the identified clusters.

The SHAP summary plot

The SHAP summary plot for yield prediction revealed the relative influence of morphometric and fruit-related descriptors in determining productivity among wood apple genotypes (Fig. 8). Tree shape (TS), fruit bearing habit (FBT), and pulp colour (PC) exerted the strongest positive impacts on yield, indicating that taller trees with favourable bearing patterns and desirable pulp traits were consistently associated with higher productivity. These findings are consistent with recent machine learning-based studies in horticultural crops, where plant architecture and fruit quality jointly explained yield variation. Leaflet-related traits, particularly lamina apex shape (LtLAS) and Leaflet lamina shape (LtLS), also contributed significantly. These descriptors likely reflect photosynthetic efficiency and assimilate partitioning, thereby supporting yield stability, in agreement with earlier observations in perennial fruit species. Spine-related traits such as spine density per meter shoot (SDPS) and spine length (SL) showed moderate effects. Their negative SHAP values at high magnitudes suggest that excessive spination may restrict canopy expansion and reduce yield, consistent with findings in arid fruit crops. Reproductive traits, including colour of open flower (COF), fruit surface texture (FST), and fruit styler end (FSyE), showed notable predictive importance. High COF and FST values were associated with improved yield, underscoring their role in pollinator attraction, reproductive success, and postharvest quality. Conversely, leaf petiole colour (LPC) had a negative impact at higher values, possibly reflecting stress-related variation. Collectively, the SHAP analysis confirmed that yield is shaped by an integrative suite of architectural, reproductive, and fruit-quality traits, emphasizing the value of explainable AI approaches for prioritizing descriptors in wood apple breeding programmes.

Comparative model performance

The comparative analysis demonstrated that the Random Forest (RF) algorithm provided the most accurate yield predictions among the tested models (Table 1). The RF model achieved a test set R^2 of 0.84, significantly outperforming the linear baseline (MLR, $R^2=0.62$) and the Support Vector Regressor (SVR, $R^2=0.76$). While the Deep Learning (MLP) model showed competitive potential ($R^2=0.71$), it failed to surpass the ensemble-based RF approach. This is likely due to the dataset size, as deep learning architectures generally require larger volumes of data to generalize effectively. The RF model also demonstrated the lowest error rates, with a Root Mean Squared Error (RMSE) of 9.45 kg and a Mean Absolute Error (MAE) of 7.12 kg, indicating a high degree of precision given the yield range of the genotypes (35–127 kg).

The random forest

The Random Forest (RF) feature importance analysis revealed considerable variation in the contribution of morphological descriptors toward yield prediction among 62 wood apple genotypes (Fig. 9). Tree shape (TS) was the most dominant predictor, contributing nearly

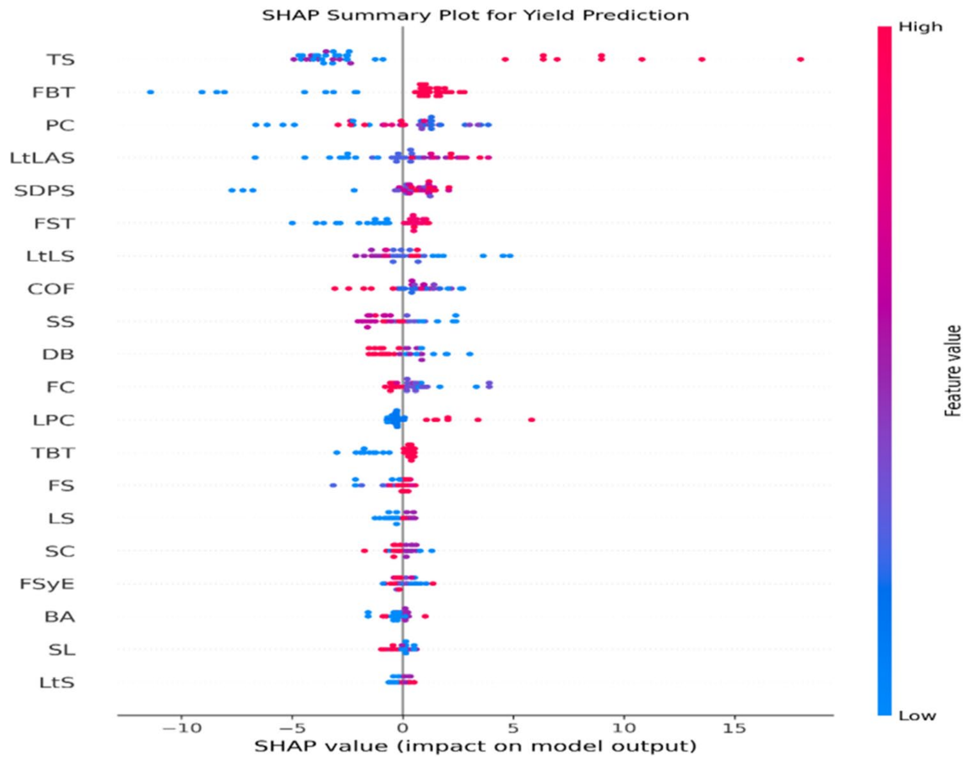

Fig. 8 SHAP summary plot for yield prediction based on morphometric descriptors of 62 wood apple genotypes. Each dot represents an individual genotype, showing the contribution of specific traits to yield prediction. The horizontal axis (SHAP value) indicates the impact of each feature on model output-positive values contribute to higher yield prediction, while negative values reduce it. The color gradient from blue to red represents the feature value, where blue indicates low trait values and red indicates high trait values. Traits such as tree shape (TS), fruit bearing habits (FBT), pulp color (PC), leaflet lamina apex shape (LtLAS), and Spine density per meter shoot (SDPS) showed strong positive SHAP values, signifying their higher influence on yield prediction. This visualization highlights the relative importance and direction of each morphological trait in determining yield potential. For abbreviations see Fig. 3

Table 1 Comparative performance metrics of different regression models for yield prediction in wood Apple (Test Set = 20%)

Model	Algorithm Type	R^2 (Test)	RMSE (kg)	MAE (kg)
Random Forest (RF)	Ensemble	0.84	9.45	7.12
Support Vector Regression (SVR)	Non-Linear	0.76	12.30	9.55
Multilayer Perceptron (MLP)	Deep Learning	0.71	14.15	11.20
Multiple Linear Regression (MLR)	Linear Baseline	0.62	16.80	13.45

16% of the total importance. This emphasizes the central role of canopy architecture in determining productivity, as a wider spread enhances light interception, photosynthetically active surface area, and carbohydrate assimilation, thereby improving yield per tree (YT). Floral and pulp-related traits also exhibited high importance. The colour of open flower (COF; 11.3%) ranked second, highlighting its role in pollinator attraction and fruit set. Pulp colour (PC; 9.0%) was the third most influential variable, reflecting its association with biochemical composition and consumer preference, while indirectly influencing yield through maturity and harvest index. Leaflet traits, particularly lamina shape (LtLAS; 7.1%) and leaflet lamina shape (LtLS; 5.0%), contributed significantly. These traits influence microclimate regulation, gas exchange, and water-use efficiency, thereby supporting yield stability under stress. Reproductive and defensive traits, including fruit-bearing habit (FBT; 5.4%) and spine density per meter shoot (SDPS; 6.0%), showed moderate contributions, the latter reflecting a balance between defense and assimilate allocation. Seed-related descriptors such as seed shape (SS) and seed colour (SC) showed intermediate importance, while fruit traits (FST, FStE, FSyE) contributed modestly, primarily to postharvest quality. In contrast, traits like NFC, LtLA, LtCV, and pedicel colour (PdC) were least important. Collectively, yield prediction was driven by an integrative suite of architectural, reproductive, and morphometric traits, supporting their prioritization in breeding programmes.

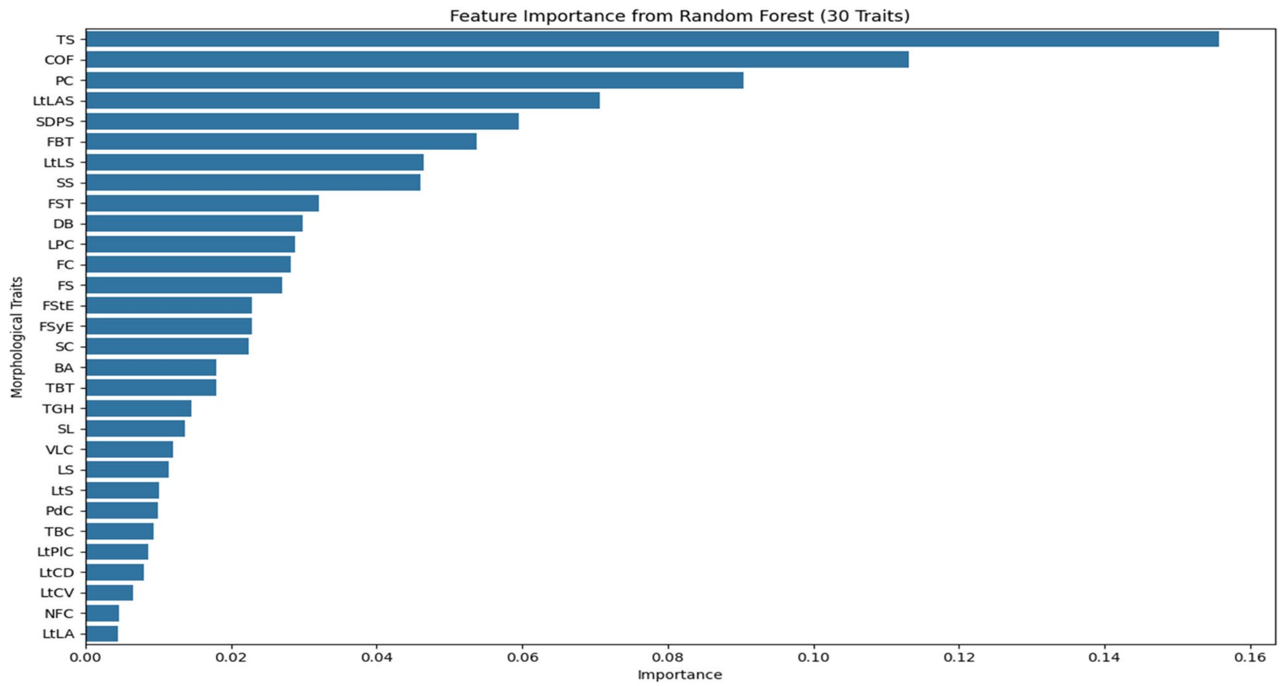

Fig. 9 Feature importance derived from the Random Forest (RF) model for discrimination among 62 wood apple genotypes based on 30 morphometric traits. The plot ranks traits according to their relative contribution to model accuracy. Tree shape (TS), colour of open flower (COF), and pulp color (PC) emerged as the most influential variables, followed by leaf, spine and fruit morphological traits such as LtLAS, SDPS, and FBT. The declining importance toward the bottom of the chart indicates that a few key traits predominantly drive genotype differentiation and yield prediction in wood apple (For abbreviations see Fig. 3.)

Discussion

Morphometric diversity and ideotype design in wood Apple

The extensive morphological variation observed among the 62 genotypes spanning canopy architecture, leaf characteristics, floral descriptors, fruit morphology, and seed traits demonstrates a broad genetic base in wood apple (Fig. 2). Such phenotypic richness is essential for ideotype-oriented breeding because it enables selection of genotypes with desirable combinations of compact canopy form, reproductive stability, and consumer-preferred fruit characteristics [46, 47]. Variation in tree shape, branch angle, and leaf geometry indicates adaptability to diverse semi-arid environments, consistent with observations in grape, guava, mango, olive and citrus where canopy traits strongly influence vigour, stress tolerance and yield stability [19, 27–29, 48].

Floral variability (colour, position, and inflorescence pattern) and fruit-related attributes (pulp colour, fruit shape, stylar-end morphology) contribute to pollinator behaviour, fruit set, and market demand [26, 49]. The reproductive plasticity observed flowers borne both terminally and laterally in clusters supports efficient cross-pollination, essential for yield improvement in cross-pollinated perennial fruit trees. Such diversity provides the foundational phenotypic space for applying multivariate and ML-driven tools (PCA, RF, SHAP) to identify superior morphotypes [50–53].

Trait–yield relationships and implications for breeding

Correlation analysis consistently identified tree shape (TS), fruit-bearing habit (FBT), pulp colour (PC), and tree-bearing tendency (TBT) as strong positive predictors of yield (Fig. 3). These traits reflect complementary aspects of vegetative vigour, source–sink balance, and reproductive efficiency, aligning with reports from date palm, guava, Indian gooseberry, tamarind and jujube that emphasize canopy architecture and leaf traits as major determinants of yield [54–58]. Pulp colour also functioned as a dual-purpose trait linked to both fruit quality and productivity reinforcing its utility as a selection criterion in ideotype development.

PCA complemented these findings by illustrating clear genotype separation driven by canopy traits (TGH, TS, BA, DB), leaf traits (LS, LtS, LtLA, LtLAS), and fruit attributes (PC, FS, SS). The first two PCs explained 31.1% of the total variation, typical for high-dimensional morphological datasets [13, 59]. Genotypes with wider branch angles, robust stature, and larger leaf area aligned with higher-yield PCA quadrants, similar to observations in grape and jujube where leaf area and canopy density enhanced photosynthetic capacity and yield [27, 58].

Hierarchical clustering further validated these relationships by grouping genotypes into three performance-based clusters. Cluster 2, defined by compact canopy structure and superior reproductive traits, exhibited the highest and most stable yields (84.6 kg $tree^{-1}$), consistent with the principle that moderate vigour combined with efficient reproductive allocation optimizes productivity [60, 61]. Cluster 0 showed high variability due to excessive vegetative growth, while Cluster 1 produced lower but uniform yields due to weaker bearing ability. Cluster 0 genotypes with vigorous growth are better suited for low-density orchards, while cluster 2 genotypes with compact canopies appear suitable for high-density planting systems, where compact growth enhances light interception and simplifies pruning and harvest [62, 63]. These findings mirror long-standing evidence in perennial horticulture that both canopy balance and reproductive capacity must be optimized simultaneously [64, 65].

Contribution and limitations of explainable machine learning in perennial fruit crops

Random Forest and SHAP analyses provided biologically interpretable insights into non-linear trait–yield relationships. RF identified TS (~16%), colour of open flower (~11%), and pulp colour (~9%) as the strongest predictors of yield (Fig. 9), in agreement with ML-based studies in guava, apple and olive where architectural and reproductive traits dominated model importance rankings [28, 48, 66]. Among machine learning approaches, Random Forest (RF) is now frequently employed as a leading tool for yield prediction [67]. SHAP values further illustrated the direction and magnitude of trait effects, highlighting synergistic contributions among canopy architecture, fruit quality traits, and reproductive features [58, 68].

Traits associated with spination (SDPS, SL) displayed negative SHAP values at high magnitudes. This aligns with ecological principles of the growth–defence tradeoff: excessive spination imposes carbon and nutrient costs that divert assimilates away from reproductive growth [58, 69, 70]. High spine density may also reduce effective leaf area index by restricting canopy expansion or causing localized shading, thereby lowering photosynthetic efficiency [59, 71]. In addition, spines can obstruct pollinator access to flowers, reducing fruit set and yield [49, 72]. These physiological mechanisms explain the consistent negative contributions of spine-related traits in SHAP outputs and reaffirm the importance of minimizing excessive spination during ideotype selection.

Comparative model performance further strengthened the robustness of the ML framework adopted in this study. The Random Forest (RF) model consistently outperformed Multiple Linear Regression (MLR), Support Vector Regression (SVR), and the Deep Learning (MLP) model, achieving the highest predictive accuracy (R^2 = 0.84) and the lowest error statistics (RMSE = 9.45 kg; MAE = 7.12 kg). The superiority of RF reflects its strong ability to model non-linear interactions and multicollinearity among morphometric traits—properties typical of complex perennial fruit systems. Similar dominance of RF over linear and kernel-based models has been reported for yield prediction [73–75]. In contrast, the weaker performance of MLR highlights the limitations of linear assumptions in biologically hierarchical trait systems [76]. Although the MLP model showed moderate predictive ability (R^2 = 0.71) in fruit crops, its inability to surpass RF is likely attributable to the limited dataset size, as deep learning architectures typically require larger training samples for stable generalization [77, 78]. Together, these results confirm that ensemble-based ML models provide an optimal balance between predictive accuracy and biological interpretability in underutilized perennial fruit crops.

The combined use of RF and SHAP thus demonstrates the practical value of explainable ML for trait prioritization, especially in underutilized species lacking formal breeding programs. However, limitations must be acknowledged. The dataset size, though adequate for tree-based ML algorithms, remains modest for deep-learning architectures. The study was conducted at a single location, constraining analysis of genotype × environment interactions. The mixed use of ordinal and quantitative trait scales also introduces interpretive constraints in PCA and RF analyses [21, 79]. Future work should therefore incorporate multi-location, multi-year datasets; integrate climatic, edaphic, and remote-sensing covariates; employ high-throughput phenotyping (e.g., UAV imaging); and combine morphometric traits with molecular markers or genomic selection tools to accelerate wood apple improvement.

Conclusion

This study presents a comprehensive integration of morphometric trait evaluation with advanced machine-learning approaches to improve yield prediction in wood apple, an underutilized yet resilient fruit tree species. The analysis of 62 genotypes revealed substantial morphological diversity, with key descriptors particularly tree shape, branch architecture, pulp colour, and floral attributes emerged as dominant regulators of yield expression. By integrating PCA, Random Forest (RF) modelling, SHAP-based interpretation, and hierarchical clustering, the study demonstrates that wood apple productivity is governed by coordinated interactions among canopy architecture, reproductive efficiency, and fruit quality rather than by any single trait alone. Importantly, the newly implemented comparative model performance analysis confirmed that the Random Forest model consistently outperformed Multiple Linear Regression, Support

Vector Regression, and the Deep Learning (MLP) model, achieving the highest predictive accuracy and the lowest error rates. This finding establishes RF as a robust and reliable modelling strategy for trait-based yield prediction in underutilized perennial fruit crops, particularly under modest sample-size conditions. The identification of compact, reproductively efficient genotypes (Cluster 2) with the highest and most stable yields further highlights their potential as elite ideotypes for targeted breeding and high-efficiency orchard systems. The convergence of conventional statistical approaches with explainable machine-learning tools not only enhances predictive power but also ensures biological interpretability of complex, non-linear trait–yield relationships. Positive contributors such as favourable canopy architecture and pulp characteristics, together with negative influences such as excessive spine density, provide actionable trait-based guidelines for selection, pruning strategies, and orchard management. Overall, this work advances the scientific understanding of the drivers of wood apple yield and establishes a scalable, explainable, and data-driven framework for precision breeding and horticultural improvement.

Abbreviations

AI	Artificial Intelligence
BA	Branch Angle
COF	Colour of Open Flower
CV	Coefficient of Variation
DB	Density of Branch
FBT	Fruit Bearing Habits
FC	Fruit Colour
FS	Fruit Shape
FST	Fruit Surface Texture
FStE	Fruit Stem End
FSyE	Fruit Styler End
LPC	Leaf Petiole Colour
LS	Leaf Size
LtCD	Leaflet Colour Dorsal side
LtCV	Leaflet Colour Ventral side
LtLA	Leaflet Lamina Attachment
LtLAS	Leaflet Lamina Apex Shape
LtLS	Leaflet Lamina Shape
LtPlC	Leaflet petiolule colour
LtS	Leaflet Size
ML	Machine Learning
NFC	New Flush Colour
PC	Principal Component
PC	Pulp Colour
PCA	Principal Component Analysis
PdC	Pedicel Colour
RCBD	Randomized Complete Block Design
RF	Random Forest
SC	Seed Colour
SDPS	Spine Density Per meter Shoot
SHAP	SHapley Additive exPlanations
SL	Spine Length
SS	Seed Shape
TBC	Tree Bark Colour
TBT	Tree Bearing Tendency
TGH	Tree Growth Habit
TS	Tree Shape
VLC	Vegetative Life Cycle
YT	Yield/ Tree

Supplementary Information

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Supplementary Material 1.

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Statement on experimental research and field studies on plants

The either cultivated or wild-growing plants sampled in accordance with relevant institutional, local, national, and international regulations.

Author contributions

VY: Conceptualization, methodology, and investigation; DSM: Methodology, Data curation, Writing original draft preparation; JR: Supervision, Conceptualization; VVA, PR, PBK, MNK, MAA: Writing-review & editing; PSK: Data analysis and visualization, Software. All authors have read and agreed to the published version of the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

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Consent for publication

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Competing interests

The authors declare no competing interests.

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References

1. Hanif T, Murad R, Amam RS, Ishrat Fatima MZ. Role of horticulture in addressing food security and global nutrition challenges. Cornous Biol. 2024;2(1):45–51. https://doi.org/10.37446/corbio/ra/2.1.2024.45-51.
2. Ahmed M, Babayola M, Bake ID. Role of horticultural crops in food and nutritional security: a review. J Nutr Food Process. 2024;7(8):01–6. https://doi.org/10.31579/2637-8914/226.

3. Qureshi AA, Kumar KE, Shaista Omer SO. *Feronia limonia*-a path less travelled. Int J Res Ayur Pharm. 2010;1:98–106.
4. Ghosh SN, Banik AK, Banik BC, Bera B, Roy S, Kundu A. Conservation, multiplication and utilization of wood Apple (*Feronia limonia*)-a semi-wild fruit crop in West Bengal (India). Acta Hortic. 2011;948:279–83. https://doi.org/10.17660/ActaHortic.2012.948.32.
5. Jassim JM, Gurudivya P, Sankar C, Raja V, Sundarrajan RV, Reddy PSK, Thamizhvanan A, Aravind S. Biochemical evaluation of diverse wood Apple (*Feronia limonia* L.) geno-types under Tamil Nadu conditions. Int J Plant Soil Sci. 2025;37(8):354–61. https://doi.org/10.9734/ijpss/2025/v37i85637.
6. Kruger RR, Navarro L. Citrus germplasm resources. In: Khan IA, editor. Citrus genetics, breeding and biotechnology. UK: CABI; 2007. pp. 45–140.
7. Srivastava R, Mishra N, Agarwal S, Mishra N. Pharmacological and phytochemical properties of Kaitha (*Feronia limonia* L.): a review. Plant Arch. 2019;19(1):608–15.
8. Sharma P, Tenguria RK. Phytochemical properties and health benefits of *Limonia acidissima*: A review. Int Res J Plant Sci. 2021;12(3):1–6.
9. Mohamed Jassim J, Gurudivya P, Sankar C, Sundarrajan RV, Anchana K, Reddy PS, Raja V, Aravind S. Evaluation of wood Apple genotypes for yield traits under Tamil Nadu condition. J Adv Biol Biotechnol. 2025;28(8):670–77. https://doi.org/10.9734/jabb/2025/v28i82741.
10. Anonymous. 2024. Director of Horticulture, Agriculture, farmers welfare and Co-operation department. Govt Gujarat. Available at https://share.google/YCISBzPEIh7c50ejL. Accessed 9 Dec 2025. https://share.google/YCISBzPEIh7c50ejL
11. Thakur N, Chugh V, Dwivedi S. Wood apple: an underutilized miracle fruit of India. Pharma Innov J. 2020;9:198–202.
12. Shukla U, Lata R, Maji S, Meena RC. Historical background, origin, distribution & present status of wood Apple. J Adv Biol Biotechnol. 2024;27(10):1457–67. https://doi.org/10.9734/jabb/2024/v27i101566.
13. Yadav V, Singh AK, Mishra DS, Yadav LP, Apparao VV, Sahil A, Ravat P, Rane J, Meena MK, Siddiqui MH, Alamri S, Khan S. Genetic diversity, quality traits, antioxidant properties, and nutrient composition of *Feronia limonia* accessions from a semiarid region for breeding and quality improvement. Sci Rep. 2025;15:35352. https://doi.org/10.1038/s41598-025-19356-1.
14. Filippi P, Jones EJ, Wimalathunge NS, Somarathna PD, Pozza LE, Ugbaje SU, Jephcott TG, Pat-erson SE, Whelan BM, Bishop TF. Precis Agric. 2019;20:1015–29. An approach to forecast grain crop yield using multi-layered, multi-farm data sets and machine learning. https://doi.org/10.1007/s11119-018-09628-4
15. Cubillas JJ, Ramos MI, Jurado JM, Feito FR. A machine learning model for early prediction of crop yield, nested in a web application in the cloud: a case study in an Olive grove in Southern Spain. Agriculture. 2022;12(9):1345. https://doi.org/10.3390/agriculture12091345.
16. Xu X, Gao P, Zhu X, Guo W, Ding J, Li C, Zhu M, Wu X. Design of an integrated Climatic as-sessment indicator (ICAI) for wheat production: A case study in Jiangsu Province, China. Ecol Indic. 2019;101:943–53. https://doi.org/10.1016/j.ecolind.2019.01.059.
17. Beulah R. A survey on different data mining techniques for crop yield prediction. Int J Comput Sci Eng. 2019;7:738–44.
18. Nasr I, Nassar K, Karray F. 2022. Enhancing fresh produce yield forecasting using vegetation indices from satellite images. IEEE International Conference on Systems, Man, and Cybernetics (SMC). 2022;2863–2868. https://doi.org/10.1109/IJCNN55064.2022.9892192
19. Suarez LA, Robson A, Brinkhoff J. Early-Season forecasting of citrus block-yield using time series remote sensing and machine learning: A case study in Australian orchards. Int J Appl Earth Obs Geoinf. 2023;122:103434. https://doi.org/10.1016/j.jag.2023.103434.
20. Jabed MA, Murad MA. Crop yield prediction in agriculture: A comprehensive review of machine learning and deep learning approaches, with insights for future research and sustainability. Heliyon. 2024;10(24). https://doi.org/10.1016/j.heliyon.2024.e40836.
21. Coşkun ÖF, Aydın A, Başak H, Mavi K, Yetişir H, Toprak S. Perspectives on morphology, physiology, genetic polymorphism and machine learning in cucumber grafting under zinc toxicity. BMC Plant Biol. 2025;25(1):1647. https://doi.org/10.1186/s12870-025-07709-x.
22. Toprak S, Coşkun ÖF. Machine learning-based evaluation of nutrient distribution in grafted cucumber plants. PhytoTalks. 2025;2(3):457–66. https://doi.org/10.21276/pt.2025.v2.i3.3.
23. Panchbhai KG, Lanjewar MG, Naik AV. Modified MobileNet with leaky ReLU and LSTM with balancing technique to classify the soil types. Earth Sci Inf. 2025;18:77. https://doi.org/10.1007/s12145-024-01521-1.
24. Panchbhai KG, Lanjewar MG. Enhancement of tea leaf diseases identification using modified SOTA models. Neural Comput Applic. 2025;37:2435–53. https://doi.org/10.1007/s00521-024-10758-2.
25. Panchbhai KG, Lanjewar MG, Malik VV, Charanarur P. Small size CNN (CAS-CNN), and modified MobileNetV2 (CAS-MODMOBNET) to identify cashew nut and fruit diseases. Multimed Tools Appl. 2024;83(42):89871–91. https://doi.org/10.1007/s11042-024-19042-w.
26. Bhandari HR, Bhanu AN, Srivastava K, Singh MN, Shreya HA. Assessment of genetic diversity in crop plants-an overview. Adv Plants Agric Res. 2017;7(3):279–86.
27. Ahmad M, Singh Z, Khan AS, Abbasi NA. Influence of canopy characteristics and leaf physiology on yield and fruit quality in grapevines (*Vitis vinifera* L). Sci Hortic. 2004;99(3–4):257–66. https://doi.org/10.1016/j.scienta.2003.08.004.
28. Mishra DS, Berwal MK, Singh A, Singh AK, Rao VA, Yadav V, Sharma BD. Phenotypic diversity for fruit quality traits and bioactive compounds in red-fleshed guava: insights from multivariate analyses and machine learning algorithms. S Afr J Bot. 2022;149:591–603. https://doi.org/10.1016/j.sajb.2022.06.043.
29. Ibell PT, Normand F, Wright CL, Mahmud K, Bally IS. The effects of planting density, training system and cultivar on vegetative growth and fruit production in young Mango (*Mangifera indica* L.) trees. Horticulturae. 2024;10(9):937. https://doi.org/10.3390/horticulturae10090937.
30. Orlenko A, Moore JH. A comparison of methods for interpreting random forest models of genetic association in the presence of non-additive interactions. BioData Min. 2021;14:9. https://doi.org/10.1186/s13040-021-00243-0.
31. Toplu C, Tunç Y, Khadivi A, Mishra DS. Integrative chemometrics, multivariate, and artificial neural network-based modeling of fruit quality, fatty acid composition, and biochemical traits in Olive (*Olea Europaea* L.) cultivars. LWT. 2025;118750. https://doi.org/10.1016/j.lwt.2025.118750.
32. Yadav V, Singh AK, Hiwale SS, Singh S, Sharma BD. Thar prabha: A new high-yielding wood Apple variety for dry land. Indian Horti. 2023;68:7–9.
33. IPGRI-International Plant Genetic Resources Institute. Descriptors for Citrus, Rome. IBPGR. 1999. http://www.bioversityinternational.org/e-library/publications/detail/descriptors-for-citrus-emcitrusem-spp/(1999).
34. Mishra DS, Singh S, Singh AK, Yadav V. Genetic variability in acid lime accessions from central Gujarat. Indian J Hort. 2018;75(4):703–8. https://doi.org/10.5958/0974-0112.2018.00117.2.
35. Singh AK, Pandey D, Kumar R, Yadav V, Yadav LP, Gangadhara K, Rane J, Kumar R, Krishna H. Descriptors for characterization and evaluation of Bael (*Aegle Marmelos* (L.) Correa ex Roxb.) germplasm for utilization in crop improvement. Genet Resour Crop Evol. 2024;71:4209–38. https://doi.org/10.1007/s10722-024-01903-w.
36. McKinney W. Data structures for statistical computing in Python. In: Proceedings of the 9th Python in Science Conference, 2010;51–56. https://doi.org/10.25080/Majora-92bf1922-00a
37. Harris CR, Millman KJ, Van Der Walt SJ, Gommers R, Virtanen P, Cournapeau D, Wieser E, Taylor J, Berg S, Smith NJ, Kern R. Array programming with numpy. Nature. 2020;585(7825):357–62. https://doi.org/10.1038/s41586-020-2649-2.
38. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, Prettenhofer P, Weiss R, Dubourg V, Vanderplas J. Scikit-learn: machine learning in python. J Mach Learn Res. 2011;12:2825–30.
39. Waskom M, Seaborn. Statistical data visualization. J Open Source Softw. 2021;6(60):3021. https://doi.org/10.21105/joss.03021.
40. Hunter JD, Matplotlib. A 2D graphics environment. Comput Sci Eng. 2007;9(3):90–5. https://doi.org/10.1109/MCSE.2007.55.
41. Lundberg SM, Lee SI. A unified approach to interpreting model predictions. Adv Neural Inf Process Syst. 2017;30:4765–74.
42. Cohen J, Cohen P, West SG, Aiken LS. Applied multiple regression/correlation analysis for the behavioral sciences. Routledge. 2013. https://doi.org/10.4324/9780203774441.
43. Jolliffe IT. Principal component analysis. 2nd ed. Springer; 2002.
44. Breiman L. Random forests. Mach Learn. 2001;45(1):5–32. https://doi.org/10.1023/A:1010933404324.
45. Hartigan JA, Wong MA, Algorithm AS. A K-means clustering algorithm. J R Stat Soc Ser C Appl Stat. 1979;136(1):100–8. https://doi.org/10.2307/2346830.
46. Mishra DS, Yadav V, Yadav LP, Appa Rao VV, Ravat P, Kumar P, Rane J. Optimizing guava (*Psidium Guajava* L.) breeding programs: the role of genotype selection in enhancing growth, yield and nutrient content. Indian J Arid Hortic. 2025;7:36–44. https://doi.org/10.48165/ijah.7.1.5.
47. Costes E, Lauri PE, Laurens F, Moutier N, Belouin A, Delort F, Legave JM, Regnard JL. Morphological and architectural traits on fruit trees which could be relevant for genetic studies: a review. Acta Hortic. 2004;663:349–55.

48. Villalobos FJ, Testi L, Hidalgo J, Pastor M, Orgaz F. Modelling potential growth and yield of Olive (*Olea Europaea* L.) canopies. Eur J Agron. 2006;24(4):296–303. https://doi.org/10.1016/j.eja.2005.10.008.
49. Ollerton J, Winfree R, Tarrant S. How many flowering plants are pollinated by animals? Oikos. 2011;120(3):321–6. https://doi.org/10.1111/j.1600-0706.2010.18644.x.
50. Eli-Chukwu NC. Applications of artificial intelligence in agriculture: A review. Eng Technol Appl Sci Res. 2019;9(4):4377–83.
51. Sheikh M, Iqra F, Ambreen H, Pravin KA, Ikra M, Chung YS. Integrating artificial intelligence and high-throughput phenotyping for crop improvement. J Integr Agric. 2024;23(6):1787–802. https://doi.org/10.1016/j.jia.2023.10.019.
52. Li L, Zhang Q, Huang D. A review of imaging techniques for plant phenotyping. Sens. 2014;14(11):20078–111. https://doi.org/10.3390/s141120078.
53. Perez-Sanz F, Navarro PJ, Egea-Cortines M. Plant phenomics: an overview of image acquisition technologies and image data analysis algorithms. Gigascience. 2017;6(11):gix092. https://doi.org/10.1093/gigascience/gix092.
54. Meena R, Chaudhary MK, Gurjar PS, Rane J, Singh AK, Yadav LP, Kumawat KL, Singh D, Chaudhary BR, Jatav MK, Singh RS. Morphological diversity assessment in date palm (*Phoenix dectylifera* L.) germplasms grown under hot arid region of India. BMC Plant Biol. 2025;25(1):1159. https://doi.org/10.1186/s12870-025-07194-2.
55. Chand R, Singh SK, Verma A. Trait associations and genetic variability in guava (*Psidium Guajava* L.) for yield and quality improvement. Sci Hortic. 2022;304:111298. https://doi.org/10.1016/j.scienta.2022.111298.
56. Singh AK, Pandey D, Gangadhara K, Yadav LP, Rane J, Krishna H, Devanand G, Pawar A, Sahil A, Ravat P. Descriptors for characterization and evaluation of Indian gooseberry (*Emblica officinalis* Gaertn) germplasm for utilization in crop improvement. Genet Resour Crop Evol. 2025;72(3):3289–319. https://doi.org/10.1007/s10722-024-02135-8.
57. Singh AK, Yadav V, Rao VA, Mishra DS, Yadav LP, Gangadhara K, Rane J, Sahil A, Ravat P, Janani P, Kaushik P. Characterization and evaluation of tamarind (*Tamarindus indica* L.) germplasm: implications for tree improvement strategies. BMC Plant Biol. 2025;25(1):396. https://doi.org/10.1186/s12870-025-06415-y.
58. Zhao X, Zhang C, Jing S, Tang Z, Lin M, Xiong R. Prediction model of jujube yield and first-order fruit rate based on BP neural network and SHAP analysis. Agronomy. 2025;15(12):2763. https://doi.org/10.3390/agronomy15122763.
59. Sankharé M, Diallo AM, Ba HS, Diatta S, Samb CO, Touré MA, Badiane S. Phenotypic diversity of growth, leaf and yield-related traits in cashew (*Anacardium occidentale* L.): implications for the development of a cashew breeding program in Senegal. Genet Resour Crop Evol. 2025;72:6771–81. https://doi.org/10.1007/s10722-025-02367-2.
60. Smith MR, Rao IM, Merchant A. Source-sink relationships in crop plants and their influence on yield development and nutritional quality. Front Plant Sci. 2018;9:1889. https://doi.org/10.3389/fpls.2018.01889.
61. Hawkesford MJ, Griffiths S. Exploiting genetic variation in nitrogen use efficiency for cereal crop improvement. Curr Opin Plant Biol. 2019;49:35–42. https://doi.org/10.1016/j.pbi.2019.05.003.
62. Simionca Mărcăşan LI, Pop R, Somsai PA, Oltean I, Popa S, Sestras AF, Militaru M, Botu M, Sestras RE. Comparative evaluation of pyrus species to identify possible resources of interest in Pear breeding. Agronomy. 2023;13(5):1264. https://doi.org/10.3390/agronomy13051264.
63. Reta K, Netzer Y, Lazarovitch N, Fait A. Canopy management practices in warm environment vineyards to improve grape yield and quality in a changing climate. A review A vademecum to vine canopy management under the challenge of global warming. Sci Hortic. 2025;341:113998. https://doi.org/10.1016/j.scienta.2025.113998.
64. Rugini E, Baldoni L, Muleo R, Sebastiani L, Mariotti R. Physiological and molecular aspects of breeding fruit tree crops for adaptation to changing climate. Hortic Rev. 2016;44:1–72.
65. Tripathi PC, Sane A, Kumar P, Chaturvedi K, Mishra DS, Ravat P. Phenotypic diversity and genetic characterization of *Cordia Myxa* L. using multivariate analysis. Flora. 2025;323:152673. https://doi.org/10.1016/j.flora.2025.152673.
66. Sharma A, Das P, Barman S, Sharma SK, Ahmed B, Adhikary T. A comparative study on the predictive ability of machine learning- and deep learning-based yield prediction model in horticulture: a case study of Apple. Appl Fruit Sci. 2025;67:191. https://doi.org/10.1007/s10341-025-01403-w.
67. Trentin C, Ampatzidis Y, Lacerda C, Shiratsuchi L. Tree crop yield Estimation and prediction using remote sensing and machine learning: A systematic review. Smart Agric Technol. 2024;9:100556. https://doi.org/10.1016/j.atech.2024.100556.
68. Danilevicz MF, Upadhyaya SR, Batley J, Bennamoun M, Bayer PE, Edwards D. Understanding plant phenotypes in crop breeding through explainable AI. Plant Biotechnol J. 2025. https://doi.org/10.1111/pbi.70208.
69. Zhao L, Yang Z, Wang C, Jin M, Duan J. Predicting de-handing point in bananas using crown morphology and interpretable machine learning. Agronomy. 2025;15(8):1880. https://doi.org/10.3390/agronomy15081880.
70. Herms DA, Mattson WJ. The dilemma of plants: to grow or defend. Q Rev Biol. 1992;67:283–335. https://doi.org/10.1086/417659.
71. Poorter H, Niklas KJ, Reich PB, Oleksyn J, Poot P, Mommer L. Biomass allocation to leaves, stems and roots: meta-analyses of interspecific variation and environmental control. New Phytol. 2012;193(1):30–50. https://doi.org/10.1111/j.1469-8137.2011.03952.x.
72. Fiorani F, Schurr U. Future scenarios for plant phenotyping. Annu Rev Plant Biol. 2013;64:267–91. https://doi.org/10.1146/annurev-arplant-050312-120137.
73. Zhang X, Wu Z, Cao C, Luo K, Qin K, Huang Y, Cao J. Design and operation of a deep-learning-based fresh tea-leaf sorting robot. Comput Electron Agric. 2023;206:107664. https://doi.org/10.1016/j.compag.2023.107664.
74. Asamoah E, Heuvelink GB, Chairi I, Bindraban PS, Logah V. Random forest machine learning for maize yield and agronomic efficiency prediction in Ghana. Heliyon. 2024;10(17). https://doi.org/10.1016/j.heliyon.2024.e37065.
75. Shawon SM, Ema FB, Mahi AK, Niha FL, Zubair HT. Crop yield prediction using machine learning: an extensive and systematic literature review. Smart Agric Technol. 2025;10:100718. https://doi.org/10.1016/j.atech.2024.100718.
76. Montgomery DC, Peck EA, Vining GG. Introduction to linear regression analysis. 5th Edition, Wiley, Hoboken, New Jersey.2012.
77. Li C, Zhao Q, Fei J, Cui L, Zhang X, Yin G. Prediction of vegetation indices series based on SWAT-ML: A case study in the Jinsha river basin. Remote Sens. 2025;17(6):958. https://doi.org/10.3390/rs17060958.
78. Saleh H, Amer E, Abuhmed T, Ali A, Al-Fuqaha A, El-Sappagh S. Computer aided progression detection model based on optimized deep LSTM ensemble model and the fusion of multivariate time series data. Sci Rep. 2023;13(1):16336. https://doi.org/10.1038/s41598-023-42796-6.
79. Aydın E, Cengiz MA, Demirsoy L, Demirsoy H. A hybrid analytical framework for cracking and some fruit quality features in sweet cherries. Horticulturae. 2025;11(6):709. https://doi.org/10.3390/horticulturae11060709.

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